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Reverse drive system for a motorized vehicle

Abstract

A motorized vehicle includes a multi-speed transmission having a plurality of forward gears configured to be selectively engaged to drive the motorized vehicle, an internal combustion engine selectively coupled to the transmission, and a reverse drive system selectively coupled to the transmission to provide torque to drive the motorized vehicle in a reverse direction. The reverse drive system includes an electric motor and a reverse drive gear train configured to be selectively coupled to the transmission. The transmission includes a neutral position in which the internal combustion engine is decoupled from the transmission and freely rotatable relative to the transmission. The reverse drive gear train is configured to be operably coupled to the transmission when the transmission is in the neutral position, such that the reverse drive system is operable to drive the motorized vehicle in the reverse direction without interference from the internal combustion engine.

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Background/Summary

BACKGROUND

(1) The present disclosure relates to motorized vehicles, such as two or three wheeled motorcycles, and more particularly to parking assist drive systems for motorized vehicles. Some motorized vehicles are relatively large and heavy. The size and weight of these vehicles may make it difficult for an operator to manually move the vehicle. As a result, these vehicles may be provided with a parking assist drive system, such as a reverse drive system, operably to propel the vehicle in a reverse direction at relatively slow speeds.

SUMMARY

(2) In one aspect, a motorized vehicle includes a multi-speed transmission including a shaft supporting a plurality of forward gears, a reverse drive system operably coupled to the multi-speed transmission, and a clutch disposed between the reverse drive system and the multi-speed transmission. The reverse drive system includes a driven gear coupled to an end of the shaft, the driven gear supported for rotation about a longitudinal axis of the shaft, and a drive gear coupled to the driven gear, the drive gear configured to be rotated by an electric motor to impart rotation on the driven gear. The clutch includes an adapter secured to the end of the shaft for co-rotation with the shaft, and a dog ring secured to the adapter for co-rotation with the adapter. The dog ring being axially movable along the adapter and having a plurality of dog teeth. The driven gear includes a plurality of dog recesses configured to be engaged by the dog teeth. The clutch is movable between a disengaged position, in which the dog ring is positioned along the adapter such that the dog teeth are spaced from the dog recesses, and an engaged position, in which the dog ring is positioned along the adapter such that the dog teeth are disposed within the dog recesses. The driven gear is freely rotatable relative to the shaft when the clutch is in the disengaged position. The driven gear is coupled for co-rotation with the shaft, and the reverse drive system is configured to rotate the shaft in a reverse direction when the clutch is in the engaged position.

(3) In another aspect, a powertrain for a motorized vehicle includes an internal combustion engine configured to drive the motorized vehicle in a forward direction, a transmission operably coupled to the internal combustion engine to receive torque from the internal combustion engine, a reverse drive system configured to drive the motorized vehicle in a reverse direction, opposite the forward direction, and a clutch. The transmission includes an input shaft configured to receive torque from the internal combustion engine, and an idler shaft having a plurality of gears configured to be meshed with the input shaft. The plurality of gears are configured to provide a plurality of selectable forward gear ratios for the motorized vehicle. The reverse drive system includes an electric motor configured to drive the motorized vehicle in the reverse direction, a drive gear coupled to the electric motor, and a driven gear configured to be operably coupled to the drive gear. The driven gear is configured to be selectively rotationally coupled to the transmission. The clutch is operably coupled to the driven gear and to the idler shaft of the transmission. The clutch is movable between a disengaged position, in which the driven gear is freely rotatable relative to the

idler shaft, and an engaged position, in which the driven gear is coupled for co-rotation with the idler shaft such that the electric motor of the reverse drive system is configured to rotate the idler shaft in a reverse direction.

(4) In yet another aspect, a motorized vehicle includes a multi-speed transmission operably coupled to a drive wheel of the motorized vehicle, the multi-speed transmission including a plurality of forward gears configured to be selectively engaged to drive the motorized vehicle at a selected gear ratio, an internal combustion engine selectively coupled to the multi-speed transmission and configured to provide torque to the multi-speed transmission to drive the motorized vehicle in a forward direction at a selected gear ratio, and a reverse drive system selectively coupled to the multi-speed transmission and configured to provide torque to the multi-speed transmission to drive the motorized vehicle in a reverse direction. The reverse drive system includes an electric motor and a reverse drive gear train configured to be selectively coupled to the multi-speed transmission to transfer torque from the electric motor to the drive wheel of the motorized vehicle. The multi-speed transmission includes a neutral position in which the internal combustion engine is decoupled from the multi-speed transmission and freely rotatable relative to the multi-speed transmission. The reverse drive gear train is configured to be operably coupled to the multi-speed transmission when the multi-speed transmission is in the neutral position, such that the reverse drive system is operable to drive the motorized vehicle in the reverse direction without interference from the internal combustion engine.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is side view of a motorized vehicle according to an embodiment of the present disclosure.
- (2) FIG. 2 is close up side view of a portion of the motorized vehicle of FIG. 1.
- (3) FIG. 3 is a perspective view showing internal components of a transmission of the motorized vehicle of FIG. 1.
- (4) FIG. 4 is a perspective view of FIG. 3, showing internal components of a reverse drive system.
- (5) FIG. 5 is a partial cross-sectional view of FIG. 4, taken along section line 5--5 in FIG. 4.
- (6) FIG. 6 is a top view of the reverse drive system FIG. 4.
- (7) FIG. 7 is a perspective view of a clutch of the reverse drive system.
- (8) FIG. 8 is a cross-sectional view of the clutch of FIG. 7, taken along section line 8--8 in FIG. 7.
- (9) FIG. 9 is an exploded perspective view of the clutch of FIG. 7.
- (10) FIG. 10 is a partial cross-sectional view of a portion of a transmission and reverse drive system according to another embodiment of the present disclosure.
- (11) FIG. 11 is a side view of a clutch of the reverse drive system according to another embodiment of the present disclosure.

DETAILED DESCRIPTION

(12) Before any embodiments of the disclosure are explained in detail, it is to be understood that the disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The disclosure is capable of other embodiments and of being practiced or of being carried out in various ways.

(13) FIG. 1 illustrates a motorized vehicle in the form of a motorcycle **20** including front and rear wheels W.sub.F, W.sub.R coupled to a frame **22**. The front wheel W.sub.F is supported by a fork **24** including a pair of fork tubes. A steering assembly **28** includes handlebars to be grasped by an operator. The steering assembly **28** is directly coupled to the fork tubes **24** to allow the operator to steer the front wheel W.sub.F via the steering assembly **28**. At least one straddle seat **32** is provided

at an exposed upper surface of the motorcycle **20** for supporting the operator (e.g., rider) and optionally a passenger. The rear wheel W.sub.R is a drive wheel that receives torque from a powertrain **30** to propel the motorcycle **20**. As will be described in greater detail herein, the powertrain **30** is adapted to propel the motorcycle **20** in a forward direction F and in a rearward direction R. Although illustrated as a motorcycle **20** having an open cockpit and no surrounding body, aspects of the present disclosure may be provided in alternate styles of vehicles including different two-wheeled vehicles, vehicles having three or more wheels such as a motorized trike, on or off road vehicles having two or more wheels, and vehicles that partially or fully encapsulate the operator.

(14) With reference to FIGS. **1** and **2**, the powertrain **30** includes a forward drive system **50**, a reverse drive system **54**, and a transmission **58** coupled to the forward and reverse drive systems **50**, **54** and to the rear wheel W.sub.R. The forward drive system **50** allows the operator to propel the motorcycle **20** in the forward direction F, and the reverse drive system **54** allows the operator to propel the motorcycle **20** in the reverse direction R. The transmission **58** is operably coupled to both the forward drive system **50** and the reverse drive system **54** to transmit torque from the drive systems **50**, **54** to the rear wheel W.sub.R. In the illustrated embodiment, the forward and reverse drive systems **50**, **54** are coupled to the transmission **58** in a parallel power transmission pathway orientation. Thus, the motorcycle **20**, which in some instances may be relatively large and heavy, does not require the operator to manually maneuver the motorcycle **20** in forward or reverse directions. Rather, the operator may utilize the forward drive system **50** to propel the motorcycle **20** in the forward direction F, and the operator may utilize the reverse drive system **54** to propel the motorcycle **20** in the reverse direction R. In some embodiments, the reverse drive system **54** is operable to propel the motorcycle **20** in the forward direction F and the reverse direction R at low speeds. Thus, the forward drive system **50** may be considered to be a high-speed or standard operating system to be used when operating the motorcycle **20** on roadways or in a traditional manner. The reverse drive system **54** may be considered to be an auxiliary or limited operating system to be used when maneuvering the motorcycle **20** in parking or other low-speed only situations.

(15) With continued reference to FIGS. **1** and **2**, the forward drive system **50** includes an internal combustion engine **62** (e.g., a gasoline burning V-twin engine). The internal combustion engine **62** generates torque to rotate the rear wheel W.sub.R in a forward direction via the transmission **58**, as will be understood by one of ordinary skill in the art. The rear wheel W.sub.R in turn drives the motorcycle **20** in the forward direction. In some constructions, the forward drive system **50** is a hybrid drive that includes the internal combustion engine **62** and at least one electric machine (e.g., a rotary motor operable with electric energy to output torque, and operable in a separate mode as a generator to convert rotational kinetic energy into electric energy for consumption and/or storage). Such an electric machine may be implemented as a starter-generator, either on or spaced from a crankshaft of the internal combustion engine **62**.

(16) With reference to FIGS. **3-6**, the transmission **58** is operably coupled to the rear wheel W.sub.R to transmit power and/or torque from the forward and reverse drive systems **50**, **54** to the rear wheel W.sub.R. The transmission **58** includes a transmission housing **66** (FIG. **2**) and is supported by the frame **22** of the motorcycle **20**. The transmission **58** of the illustrated embodiment is a multi-speed transmission having multiple gear pairs **70** disposed on parallel transmission shafts, illustrated as an input shaft **74** and an idler shaft **78**, located within the transmission housing **66**. For example, a transmission **58** may include six gear pairs **70** defining six distinct forward gear ratios. The transmission **58** does not include a reverse gear ratio. Within each gear pair **70**, an input gear **82** is disposed on the input shaft **74** and an idler shaft gear **86** is disposed on the idler shaft **78** (FIG. **6**). The input gear **82** is coupled for co-rotation with the input shaft **74** and is meshed with the idler shaft gear **86**. The idler shaft gear **86** is disposed on the idler shaft **78** and is selectively securable for co-rotation with the idler shaft **78**. Thus, when the idler shaft gear **86** is not secured

for co-rotation with the idler shaft **78**, the idler shaft gear **86** is freely rotatable relative to the idler shaft **78**. The transmission **58** is operable such that, of the gear pairs **70**, only one idler shaft gear **86** is capable of being secured for co-rotation with the idler shaft **78** at a time. The remainder of the idler shaft gears **86** are freely rotatable relative to the idler shaft **78** and do not transmit torque to the idler shaft **78**. In some embodiments, the transmission **58** further includes a bottle gear **90** (FIG. **4**) positioned coaxially with the input shaft **74** and freely rotatable relative to the input shaft **74**. The idler shaft **78** is permanently meshed with the bottle gear **90**, and the bottle gear functions as an output of the transmission **58**. In other embodiments, the transmission **58** may not include a bottle gear, and the idler shaft **78** may function as the output **90** of the transmission **58**. As will be understood by one of ordinary skill in the art, the output **90** of the transmission **58** is operably coupled to the rear wheel W.sub.R to transmit power and/or torque to the rear wheel W.sub.R (e.g., via a belt or chain drive). The input shaft **74** is coupled to the forward drive system **50** to receive power and/or torque from the internal combustion engine **62**. For example, a primary drive (not shown) may couple the input shaft **74** of the transmission **58** to the internal combustion engine **62**, as will be understood by one of ordinary skill in the art. The transmission **58** further includes a gear setting or selection mechanism, illustrated as a shift drum **94**, that is controllable by the operator to select a desired one of the gear pairs **70** (e.g., a desired forward gear ratio). The gear selection mechanism **94**, and thus the transmission **58**, is also capable of a neutral position in which no gear pair **70** is selected. When in the neutral position, no idler shaft gear **86** is secured for co-rotation with the idler shaft **78**, such that the idler shaft **78** is freely rotatable relative to the input shaft **74** and torque cannot be transmitted from the internal combustion engine **62** to the rear wheel W.sub.R. The idler shaft **78** maintains its coupling with the output **90** of the transmission **58** when the transmission **58** is in neutral. In the illustrated embodiment, the gear selection mechanism **94** is manually or electro-mechanically operable such that the transmission **58** functions as a standard or manual transmission. While a transmission **58** having 6 gear pairs is described herein, it should be understood that in other constructions the transmission may have more or fewer gear pairs defining more or fewer forward gear ratios as desired. For example, a transmission may include 3, 4, 5, or 7 forward gear pairs.

(17) With reference to FIGS. **3-6**, the reverse drive system **54** includes an electric motor **98** that generates torque to rotate the rear wheel W.sub.R in the reverse direction which, in turn, propels the motorcycle **20** in the reverse direction R. The reverse drive system **54** further includes a reverse drive gear train **102** to transmit torque from the electric motor **98** to the transmission **58** which, in turn, transmits the torque from the electric motor **98** to the rear wheel W.sub.R. In the illustrated embodiment, the reverse drive gear train **102** is disposed within a reverse drive housing **106**, located outside of the transmission housing **66**, and includes a plurality of gears configured to multiply torque from the electric motor **98**. The reverse drive housing **106** is coupled to the transmission housing **66**. In some embodiments, the reverse drive system **54** is included on the motorcycle **20** when assembled by the manufacturer. In other embodiments, the reverse drive system **54** may be retrofitted onto the motorcycle **20**.

(18) With continued reference to FIGS. **3-6**, the reverse drive gear train **102** includes at least a drive gear **110** coupled to the electric motor **98** and a driven gear **114** coupled to the drive gear **110**. The drive gear **110** is meshed with an output pinion **100** of the electric motor **98** to receive torque therefrom. In the illustrated embodiment, the drive gear **110** includes a torque limiter **112** that allows the drive gear **110** to “slip” via a set of clutch plates or other mechanism when an amount of torque to be transferred is above a set value. The torque limiter **112** may prevent back driving of the electric motor **98** or other damage to the reverse drive system **54**. The driven gear **114** is also coupled to the transmission **58** via the idler shaft **78** and is selectively securable for co-rotation with the idler shaft **78** to transmit torque from the electric motor **98** to the transmission **58**. The reverse drive gear train **102** of the illustrated embodiment includes the drive gear **110**, an idler gear **118** meshed with the drive gear **110**, and the driven gear **114** meshed with the idler gear **118**. The drive

gear **110**, the idler gear **118**, and the driven gear **114** are permanently meshed. The drive gear **110** is a compound gear supported by a drive gear shaft **122** and including a drive gear input portion **110a** and a drive gear output portion **110b**. The drive gear input portion **110a** is relatively larger than the drive gear output portion **110b** (e.g., the drive gear input portion **110a** has a higher tooth count than the drive gear output portion **110b**) and is meshed with the electric motor **98**. In other constructions, the drive gear input portion **110a** may be relatively smaller than the drive gear output portion **110b**, depending on a desired gear ratio between the electric motor **98** and the transmission **58**. The drive gear output portion **110b** is secured for co-rotation with the drive gear input portion **110a** and is meshed with the idler gear **118**. The drive gear input portion **110a** and the drive gear output portion **110b** may be separately formed gears and coupled for co-rotation via the drive gear shaft **122** to define the compound drive gear **110**. In other constructions, the drive gear input portion **110a** and the drive gear output portion **110b** may be integrally formed and/or connected to one another as a single gear having differently sized portions. The idler gear **118** is also a compound gear and is supported by an idler gear shaft **126**. The idler gear **118** thus includes an idler gear input portion **118a** and an idler gear output portion **118b**. The idler gear input portion **118a** is relatively larger than the idler gear output portion **118b** (e.g., the idler gear input portion **118a** has a higher tooth count than the idler gear output portion **118b**), and the idler gear input portion **118a** is meshed with the drive gear output portion **110b**. In other constructions, the idler gear input portion **118a** may be relatively smaller than the idler gear output portion **118b**, depending on a design gear ratio between the electric motor **98** and the transmission **58**. Similar to the drive gear **110**, the idler gear input portion **118a** and the idler gear output portion **118b** may be formed as separate gears that are secured for co-rotation with the idler gear shaft **126** to form a compound gear, or the idler gear input portion **118a** and the idler gear output portion **118b** may be integrally formed as a single compound gear. The idler gear output portion **118b** is meshed with the driven gear **114**, which is coupled to the transmission **58** and selectively securable for co-rotation with the idler shaft **78**. In the illustrated embodiment, the drive gear **110**, the idler gear **118**, and the driven gear **114** remain permanently meshed with one another. The driven gear **114** is positioned to be co-rotatable with the idler shaft **78** of the transmission **58** and is selectively securable for co-rotation with the idler shaft **78**. It should be understood that, in other constructions, the reverse drive gear train **102** may include more or fewer gears and the gears may or may not be formed as compound gears based on a desired magnitude of torque multiplication (e.g., gear ratio) between the electric motor **98** and the transmission **58**.

(19) With reference to FIGS. 4-9, a reverse drive clutch **130** is disposed between the driven gear **114** and the idler shaft **78** to selectively couple the driven gear **114** for co-rotation with the idler shaft **78** and allow for the transmission of torque to the transmission **58**. The clutch **130** is actuatable between a disengaged position, in which the driven gear **114** is rotatable relative to the idler shaft **78** and an engaged position in which the driven gear **114** is secured for co-rotation with the idler shaft **78** such that the reverse drive system **54** is capable of transmitting torque to the rear wheel W.sub.R. In some constructions, the clutch **130** may be biased toward the disengaged position. The clutch **130** includes an adapter **134** coupled to the idler shaft **78**, a dog ring **138** coupled to the adapter **134**, and a shift fork **142** coupled to the dog ring **138**. The adapter **134** is secured for co-rotation with the idler shaft **78**. The dog ring **138** selectively engages the driven gear **114** to couple the clutch **130** to the reverse drive system **54** (e.g., when the clutch **130** is in the engaged position). The dog ring **138** encircles a portion of the adapter **134**, is secured for co-rotation with the adapter **134**, and is axially movable along a portion of the adapter **134**. As will be described in greater detail herein, the dog ring **138** is movable along the adapter **134** as the clutch **130** moves between the engaged position and the disengaged position.

(20) With reference to FIG. 5, an end of the idler shaft **78** extends beyond the transmission housing **66** to be coupled to the adapter **134**. The end of the idler shaft **78** includes a plurality of splines **146**. The adapter **134** is generally cylindrical and includes a central aperture **150** extending therethrough.

The central aperture **150** includes a splined portion **154** engageable with the splines **146** on the end of the idler shaft **78** to rotatably couple the adapter **134** and the idler shaft **78**. The central aperture **150** further includes a shoulder **158** engageable by a fastener **162** to axially secure the adapter **134** to the end of the idler shaft **78**. The end of the idler shaft **78** includes a threaded bore **166**, and the fastener **162** is a bolt that threadably engages the bore **166**. In the illustrated embodiment, a washer **170** is disposed about the bolt **162** to engage the shoulder **158**, such that tightening of the bolt **162** secures the adapter **134** to the end of the idler shaft **78**.

(21) With continued reference to FIGS. 5, 8, and 9, the adapter **134** includes a dog ring support portion **174** and a driven gear support portion **178**. The driven gear support portion **178** is disposed closer to the transmission housing **66** than the dog ring support portion **174** when the adapter **134** is coupled to the idler shaft **78**. The dog ring support portion **174** is splined, such that the dog ring **138** can be rotationally coupled to the adapter **134**. The driven gear support portion **178** is adapted to rotatably support the driven gear **114**. In the illustrated embodiment, a bearing assembly **182** is coupled to the driven gear support portion **178** to support the driven gear **114** for rotation relative to the adapter **134**, and the driven gear support portion **178** includes a circumferential recess **186** to locate the bearing assembly **182**.

(22) With continued reference to FIGS. 5, 8, and 9, the dog ring **138** is coupled to the adapter **134** via the dog ring support portion **174**. An inner surface **190** of the dog ring **138** is splined to engage with the splined dog ring support portion **174**. Thus, the dog ring **138** is secured to the adapter **134** for co-rotation with the adapter **134** while being axially movable relative to the adapter **134**. In the illustrated embodiment, the dog ring support portion **174** includes a stop portion **194** extending radially from an outer end thereof. The stop portion **194** functions to limit the axial extent to which the dog ring **138** is movable along the adapter **134** in a direction away from the driven gear **114**. The dog ring **138** further includes a plurality of dog teeth **198** disposed circumferentially about the dog ring **138** and extending axially away from the dog ring **138**. The driven gear **114** includes a plurality of dog recesses **202** that are selectively engageable by the dog teeth **198** to rotationally couple the driven gear **114** and the dog ring **138**. The dog ring **138** also includes a circumferential recess **206** that is engageable by the shift fork **142**. The shift fork **142** is adapted to control a position of the dog ring **138** along the dog ring support portion **174** of the adapter **134**. In some embodiments, the shift fork **142** is coupled to the shift drum **94** to be actuated (FIG. 4). In other embodiments, the shift fork **142** is coupled to a dedicated actuator, a linear actuator or solenoid **200** (FIG. 6), to be moved along the length of the adapter **134**. The clutch **130** is in the engaged position when the shift fork **142** has moved the dog ring **138** towards the driven gear **114** such that the dog teeth **198** are engaged with the dog recesses **202**, and the clutch **130** is in the disengaged position when the shift fork **142** has moved the dog ring **138** away from the driven gear **114** such that the dog teeth **198** are disengaged from the dog recesses **202**. The reverse drive system **54** cannot impart rotation on the rear wheel W.sub.R when the clutch **130** is in the disengaged position.

(23) FIG. 10 illustrates another embodiment of a reverse drive clutch **1130** with following differences explained below. Rather than an adapter **134** supporting a movable dog ring **138**, the reverse drive clutch **1130** includes an adapter, illustrated as a shaft end ratchet nut **1134**, coupled to the idler shaft **78**. The shaft end ratchet nut **1134** is threaded to an end of the idler shaft **78** in place of the splines **146**. The shaft end ratchet nut **1134** rotatably supports the driven gear **114** via a bearing assembly **182**, such that the driven gear **114** is rotatable relative to the shaft end ratchet nut **1134** and axially secured relative to the shaft end ratchet nut **1134**. A ratchet coupler **1138** is supported within driven gear **114** for co-rotation with the driven gear **114** and translational movement relative to the shaft end ratchet nut **1134**. In the illustrated embodiment, a splined connection exists between the driven gear **114** and the ratchet coupler **1138**. The ratchet coupler **1138** is movable relative to the shaft end ratchet nut **1134** between an engaged position, in which the coupler **1138** is engaged with the nut **1134** to transmit torque to the nut **1134** and thus to the idler shaft **78**, and a disengaged position, in which the ratchet coupler **1138** is spaced from the shaft

end ratchet nut **1134** to that torque cannot be transmitted from the driven gear **114** to the idler shaft **78**. In the illustrated embodiment, a biasing mechanism **1142** biases the ratchet coupler **1138** toward the disengaged position.

(24) Each embodiment of the reverse drive clutch **130**, **1130** as described herein selectively couples the driven gear **114** to the idler shaft **78** to transmit torque to the idler shaft **78**. Similarly, each embodiment utilizes axial movement (e.g., of the dog ring **138** or the ratchet coupler **1138**) to engage or disengage the clutch **130**, **1130**. However, the dog ring **130** is rotationally secured to the adapter **134** and axially movable relative to the adapter **134**, while the ratchet coupler **1138** is rotationally secured to the driven gear **114** and axially movable relative to the driven gear **114**. Thus, the component of the reverse drive clutches **130**, **1130** that is movable is oppositely contained in the two embodiments.

(25) FIG. **11** illustrates yet another embodiment of a reverse drive clutch **2130**, with the following differences explained below. Rather than the mechanical clutch embodiments described above, the reverse drive clutch **2130** is electromagnetically actuated. The electromagnetic reverse drive clutch **2130** includes, among other things, a power source **2134**, a coil assembly **2138** operably coupled to the power source, and a magnetic clutch element **2142** (e.g., a plurality of magnetic rollers) movably supported within the reverse drive clutch **2130**. The magnetic clutch element **2142** is movable (e.g., radially) between a disengaged position, in which the magnetic clutch element **2142** is disengaged from the driven gear **114** to inhibit torque transmission, and an engaged position, in which the magnetic clutch element **2142** is engaged with the driven gear **114** to transmit torque. Movement of the magnetic clutch element **2142** is induced via the power source **2134** and the coil assembly **2138**. More particularly, the power source **2134** is configured to selectively provide electric power to the coil assembly **2138** which, in turn, generates a magnetic field that acts on the magnetic clutch element **2142** to move the magnetic clutch element **2142** towards the engaged position. In some embodiments, the magnetic clutch element **2142** may be biased towards the disengaged position. The electromagnetic reverse drive clutch **2130** is disposed between the driven gear **114** and the idler shaft **78** and, when in the disengaged position, at least a portion of the electromagnetic reverse drive clutch **2130** is rotatable relative to the idler shaft **78** or the driven gear **114**.

(26) Referring again to FIG. **1**, the powertrain **30** described herein includes two parallel power transmission pathways beginning with the forward and reverse drive systems **50**, **54**, respectively. A forward drive pathway begins with the internal combustion engine **62** which generates torque to propel the motorcycle **20** in the forward direction F. The torque from the internal combustion engine **62** is then transmitted to the input shaft **74** of the multi-speed transmission, through a selected gear pair **70**, and to the output **90** of the transmission **58** to the rear wheel W.sub.R to rotate the rear wheel W.sub.R in a forward direction. A reverse drive path begins at the electric motor **98** which generates torque to propel the motorcycle **20** in the reverse direction R. The torque from the electric motor **98** is transmitted through the permanently meshed gears **110**, **118**, **114** of the reverse drive gear train **102** and to the idler shaft **78** of the transmission **58** to be outputted to the rear wheel W.sub.R to drive the rear wheel W.sub.R in the reverse direction. The reverse drive clutch **130** is in the engaged position to transmit the torque from the electric motor **98** to the rear wheel W.sub.R. The forward drive system **50** and the reverse drive system **54** are prevented from operating simultaneously so as not to damage the motorcycle **20**. To operate the forward drive system **50**, the reverse drive clutch **130** is disengaged so that the idler shaft **78** of the transmission **58** is freely rotatable relative to the reverse drive gear train **102**. Thus, the forward drive system **50** is operable without inadvertently transmitting torque to the reverse drive system **54** which could damage the reverse drive system **54**. To operate the reverse drive system **54**, the transmission **58** is placed in the neutral position to allow the idler shaft **78** to rotate relative to the input shaft **74**, and the reverse drive clutch **130** is in the engaged position to operably couple the reverse drive system **54** to the idler shaft **78** of the transmission **58**. When the transmission **58** is in neutral, the reverse

drive system **54** is operable to transmit torque to the rear wheel W.sub.R without inadvertently transmitting torque to the internal combustion engine **62** which could damage the internal combustion engine **62** or the reverse drive system **54**.

(27) In some embodiments, the electric motor **98** of the reverse drive system **54** functions as both a starter motor for the internal combustion engine **62** and to provide torque for the reverse drive system **54**. The electric motor **98** includes a reverse drive end **218** coupled to the reverse drive gear train **102** and a starter end **222** operably coupled to the internal combustion engine **62**. The starter end **222** is selectively coupled to the internal combustion engine **62** to rotate the internal combustion engine **62** during a starting operation. In the illustrated embodiment, the starter end **222** includes a bendix style output (not shown) that engages the internal combustion engine **62** during starting and disengages from the internal combustion engine **62** when not starting the internal combustion engine **62**. Thus, the starter end **222** is disengaged from the internal combustion engine **62** when the electric motor **98** provides torque for the reverse drive system **54** and does not inadvertently transmit torque to the internal combustion engine **62**. Similarly, when starting the internal combustion engine **62**, the reverse drive clutch **130** is in the disengaged position to prevent torque from the electric motor **98** to be transmitted to the transmission **58** via the reverse drive system **54**.

(28) In the illustrated embodiments, the electric motor **98** may be rotatable in forward and reverse directions. As previously mentioned, it is therefore possible to propel the motorcycle **20** in the forward direction F via the reverse drive system **54**. In such an embodiment, the reverse drive system **54** may be considered the auxiliary (e.g., low-speed) operating system and the forward drive system **50** may be considered the standard (e.g., high-speed) operating system. The low-speed operating system allows the operator to maneuver the motorcycle **20** under electric power and at low speeds during parking or other maneuvers. For example, the low-speed operating system may propel the motorcycle **20** in forward or reverse directions at a relatively low rate of speed, in comparison to the high-speed operating system. The low-speed operating system is driven by the electric motor **98** via the reverse drive gear train **102**. The low-speed operating system requires the transmission **58** to be in neutral. Therefore, the internal combustion engine **62** may be on or off during operation of the low-speed operating system, without interfering with the low-speed operating system. If the internal combustion engine **62** is on during operation of the low-speed operating system, the internal combustion engine **62** may charge a battery (not shown) that provides power to the electric motor **98** to drive the low-speed operating system thereby extending an amount of time that the low-speed operating system may be used for. The high-speed operating system allows the operator to maneuver the motorcycle **20** at standard operating speeds, for example, when on roadways.

(29) The embodiments described above and illustrated in the figures are presented by way of example only and are not intended as a limitation upon the concepts and principles of the present invention. As such, it will be appreciated by one having ordinary skill in the art that various changes in the elements and their configuration and arrangement are possible without departing from the spirit and scope of the present invention. For example, one having ordinary skill in the art will appreciate that specific features of the numerous embodiments disclosed may be mixed and matched in other ways where not specifically inhibited, even though specific illustration of such embodiments may not be exhaustively covered herein.

(30) Various features and advantages of the disclosure are set forth in the following claims.

Claims

1. A powertrain for a motorized vehicle including an internal combustion engine configured to drive the motorized vehicle in a forward direction, the powertrain comprising: a transmission operably coupled to the internal combustion engine to receive torque from the internal combustion

engine, the transmission including: an input shaft configured to receive torque from the internal combustion engine, and an idler shaft having a plurality of gears configured to be meshed with the input shaft, the plurality of gears configured to provide a plurality of selectable forward gear ratios for the motorized vehicle; a reverse drive system configured to drive the motorized vehicle in a reverse direction, opposite the forward direction, the reverse drive system including: an electric motor configured to drive the motorized vehicle in a reverse direction, a drive gear coupled to the electric motor, and a driven gear configured to be operably coupled to the drive gear, the driven gear configured to be selectively rotationally coupled to the transmission; and a clutch operably coupled to the driven gear and to the idler shaft of the transmission. the clutch movable between a disengaged position, in which the driven gear is freely rotatable relative to the idler shaft, and an engaged position, in which the driven gear is coupled for co- rotation with the idler shaft such that the electric motor of the reverse drive system is configured to rotate the idler shaft in a reverse direction, wherein the reverse drive system includes an idler gear disposed between the drive gear and the driven gear, the idler gear configured to be permanently meshed with the drive gear and the driven gear, wherein the drive gear is a compound gear having a drive gear input portion and a drive gear output portion, and wherein the drive gear input portion is relatively larger than the drive gear output portion, and wherein the multi-speed transmission includes a transmission housing in which the plurality of forward gears are disposed, and wherein the reverse drive system is disposed outside of the transmission housing.

2. The powertrain of claim 1, wherein the idler gear is a compound gear having an idler gear input portion and an idler gear output portion, and wherein the idler gear input portion is relatively larger than the idler gear output portion.

3. The powertrain of claim 1, wherein the electric motor is configured as a starter motor to start the internal combustion engine.

4. The powertrain of claim 1, wherein the clutch includes: an adapter coupled to the idler shaft, and a dog ring disposed about the adapter and movable relative to the adapter, the dog ring including a plurality of dog teeth configured to couple the driven gear for co- rotation with the dog ring.

5. The powertrain of claim 4, wherein the dog ring is axially movable relative to the adapter, and wherein the dog ring is coupled for co-rotation with the adapter.

6. A motorized vehicle comprising: a multi-speed transmission operably coupled to a drive wheel of the motorized vehicle, the multi-speed transmission including a plurality of forward gears configured to be selectively engaged to drive the motorized vehicle at a selected gear ratio; an internal combustion engine selectively coupled to the multi-speed transmission and configured to provide torque to the multi-speed transmission to drive the motorized vehicle in a forward direction at the selected gear ratio; and a reverse drive system selectively coupled to the multi-speed transmission and configured to provide torque to the multi-speed to the multi-speed transmission to drive the motorized vehicle in a reverse direction, the reverse drive system including: an electric motor, and a reverse drive gear train configured to be selectively coupled to the multi- speed transmission to transfer torque from the electric motor to the drive wheel of the motorized vehicle, the reverse drive gear train including a drive gear coupled to the electric motor, a driven gear, and an idler gear disposed between the drive gear and the driven gear, the idler gear configured to be permanently meshed with the drive gear and the driven gear, wherein the drive gear is a compound gear having a drive gear input portion and a drive gear output portion, and wherein the drive gear input portion is relatively larger than the drive gear output portion, wherein the multi-speed transmission includes a neutral position in which the internal combustion engine is decoupled from the multi-speed transmission and freely rotatable relative to the multi-speed transmission, and wherein the reverse drive gear train is configured to be operably coupled to the multi- speed transmission when the multi-speed transmission is in the neutral position, such that the reverse drive system is operable to drive the motorized vehicle in a reverse direction without interference from the internal combustion engine, and wherein the multi-speed transmission includes a

transmission housing in which the plurality of forward gears are disposed, and wherein the reverse drive system is disposed outside of the transmission housing.

7. The motorized vehicle of claim 6, wherein the reverse drive system is disposed within a reverse drive housing, and wherein the reverse drive housing is coupled to the transmission housing.

8. The motorized vehicle of claim 6, further comprising a clutch disposed between the reverse drive gear train and the multi-speed transmission.

9. The motorized vehicle of claim 6, wherein the multi-speed transmission includes an input shaft and an idler shaft, wherein the idler shaft is freely rotatable relative to the input shaft when the multi-speed transmission is in the neutral position.

10. The motorized vehicle of claim 9, wherein the reverse drive gear train is configured to be selectively coupled to the idler shaft to transfer torque from the electric motor to the drive wheel of the motorized vehicle.
